**PHYS 210 Electricity and Magnetism**

**Celebration of Knowledge!**

Supplementary Practice 5 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

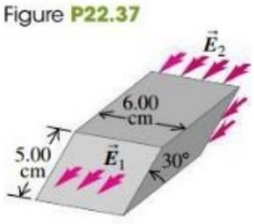
Original source numbering is retained inside each question. Answer the entire cropped question.

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**1. [____]** Homework 4 - Question 22.37

**22.37** •• The electric field $\vec{E}_1$ at one face of a parallelepiped is uniform over the entire face and is directed out of the face. At the opposite face, the electric field $\vec{E}_2$ is also uniform over the entire face and is directed into that face (**Fig. P22.37**). The two faces in question are inclined at 30.0° from the horizontal, while both $\vec{E}_1$ and $\vec{E}_2$ are horizontal; $\vec{E}_1$ has a magnitude of $2.50 \times 10^4$ N/C, and $\vec{E}_2$ has a magnitude of $7.00 \times 10^4$ N/C. (a) Assuming that no other electric field lines cross the surfaces of the parallelepiped, determine the net charge contained within. (b) Is the electric field produced by the charges only within the parallelepiped, or is the field also due to charges outside the parallelepiped? How can you tell?

Figure P22.37




**2. [____]** Tutorial 4 - Question 1

1. Two infinite, uniformly charged, flat surfaces are mutually perpendicular. One of the sheets has a charge density of +60 pC/m$^2$, and the other carries a charge density of –80 pC/m$^2$. What is the magnitude of the electric field at any point not on either surface?
   a. 1.1 N/C
   b. 5.6 N/C
   c. 7.9 N/C
   d. 3.8 N/C
   e. 4.0 N/C

**3. [____]** Tutorial 4 - Question 2

2. A long nonconducting cylinder (radius = 12 cm) has a charge of uniform density (5.0 nC/m$^3$) distributed throughout its column. Determine the magnitude of the electric field 5.0 cm from the axis of the cylinder.
   a. 25 N/C
   b. 20 N/C
   c. 14 N/C
   d. 31 N/C
   e. 34 N/C

**4. [____]** Tutorial 4 - Question 3

3. Each 2.0-m length of a long cylinder (radius = 4.0 mm) has a charge of 4.0 nC distributed uniformly throughout its volume. What is the magnitude of the electric field at a point 5.0 mm from the axis of the cylinder?
   a. 9.9 kN/C
   b. 8.1 kN/C
   c. 9.0 kN/C
   d. 7.2 kN/C
   e. 18 kN/C